Arista Access Node HSR-ACC-69 Hardware/Environmental Fault SOP (ALARM_5226)

Node: HSR-ACC-69 | **Vendor:** Arista | **Severity:** Critical | **Alarm Code:** ALARM_5226

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a critical hardware/environmental fault on Arista Access Node HSR-ACC-69, triggered by alarm code ALARM_5226. The failure domain is attributed to ASIC memory parity errors, line card crashes, and loss of power supply redundancy. On Arista Access nodes, the control plane and data plane are integrated into a single ASIC, which executes the Arista Extensible Operating System (AEOps). The ASIC is responsible for processing forwarding decisions, implementing Quality of Service (QoS) policies, and managing the node's network interfaces. In the event of an ASIC memory parity error or line card crash, the control plane may become unavailable, leading to packet loss and potential network instability. Furthermore, the loss of power supply redundancy can cause the node to shut down, resulting in a complete loss of network connectivity. The Arista Access node's redundant power supplies are designed to provide automatic failover in the event of a primary power supply failure. However, if both power supplies fail, the node will shut down, triggering the ALARM_5226 alarm. To mitigate this fault, it is essential to identify the root cause and perform targeted repairs or replacements to restore network connectivity.

Pre-requisite Checks

- Verify that the node is in a stable state and not experiencing any other critical alarms.

- Check the power supply status and ensure that both power supplies are online and functioning correctly.

- Verify that the redundant link is online and functioning correctly.

Potential Root Causes

Cause	Probability
ASIC Memory Parity Error	High
Line Card Crash	Medium
Power Supply Redundancy Loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system, Expected Output: System Information, Description: This command displays the system information, including the node's serial number, software version, and hardware configuration. The output will help identify any issues with the node's hardware or software configuration.
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```
Command: show power supply, Expected Output: Power Supply Status, Description: This command displays the power supply status, including the power supply unit (PSU) status, voltage, and current. The output will help identify any issues with the power supply.

Command: show interface, Expected Output: Interface Status, Description: This command displays the interface status, including the interface name, status, and IP address. The output will help identify any issues with the node's network interfaces.
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Resolution Steps

Step 1: Step: Power Cycle the Node, Command: reboot, Description: Perform a power cycle on the node to reset the system and clear any transient issues. Ensure that the power supplies are online and functioning correctly before proceeding.

Step 2: Step: Replace the Faulty ASIC, Command: replace asic, Description: Replace the faulty ASIC with a new one to resolve the ASIC memory parity error or line card crash issue.

Step 3: Step: Replace the Faulty Power Supply, Command: replace power supply, Description: Replace the faulty power supply with a new one to resolve the power supply redundancy loss issue.

Post-Check Verification

```
Command: show system, Expected Output: System Information, Description: Verify that the node is in a stable state and the system information is correct.

Command: show power supply, Expected Output: Power Supply Status, Description: Verify that the power supply status is correct and both power supplies are online and functioning correctly.

Command: show interface, Expected Output: Interface Status, Description: Verify that the interface status is correct and all interfaces are online and functioning correctly.
```

Roll-back Procedures

- Step: Restore the Original Configuration, Command: load configuration, Description: Restore the original configuration to the node to roll back any changes made during the resolution process.

- Step: Replace the Faulty Component, Command: replace [component], Description: Replace the faulty component with the original component to roll back any changes made during the resolution process.